

ESP32 Digital Signature Benchmark

Phase B: Integrated Network Handshake Benchmark (Over Real Wi-Fi)

FYP Documentation: Embedded VPN Client

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1 Phase B Overview: Live Network Handshake Benchmark

While Phase A evaluated raw cryptographic operations directly in ESP32 chip memory, Phase B evaluates the full 3-way mutual authentication handshake executed over real local Wi-Fi and TCP sockets connecting an ESP32 client to a laptop gateway server.

This benchmark captures real-world embedded network characteristics:

- TCP socket transport delays over IEEE 802.11 b/g/n wireless connections.
- End-to-end handshake latency including message transmission, signature generation, and verification on both sides.
- Over-the-air byte footprint and its impact on radio transmission stability.
- Latency predictability and jitter across multiple consecutive connection attempts.

Three cryptographic algorithms were benchmarked under identical network conditions:

1. **Ed25519:** Twisted Edwards Curve25519 with Libsodium on the ESP32 and Python cryptography on the server.
2. **ECDSA P-256:** NIST prime curve secp256r1 with ESP32 mbedTLS and Python cryptography.
3. **RSA-3072:** Classical 3072-bit RSA with PKCS#1 v1.5 padding and SHA-256 at an equivalent 128-bit security level.

2 Key Performance Indicators (KPIs)

The Phase B evaluation profiles five network and system performance metrics:

- **NET-1 (Total Handshake Latency):** Total elapsed time in milliseconds from the transmission of the initial server challenge to the completion of mutual verification on both ends.
- **NET-2 (Wire Overhead per Handshake):** The exact number of bytes exchanged across the TCP socket during the 3-way authentication sequence (Message 1 + Message 2 + Message 3).
- **NET-3 (Handshake Success Rate):** The percentage of initiated handshakes that successfully verified digital signatures without timing out or dropping packets.
- **NET-4 (Latency Jitter / Standard Deviation):** The variability in handshake duration across consecutive runs, indicating connection predictability.
- **NET-5 (Radio Channel Occupancy):** Qualitative impact of packet size on channel airtime and battery efficiency.

Table 1: Summary of Phase B Benchmarking KPIs

KPI ID	Metric Name	Operational Significance
NET-1	Total Handshake Latency	Real-world connection setup delay experienced by the IoT device
NET-2	Total Wire Payload	Volume of TCP authentication traffic transmitted over the radio
NET-3	Success Rate	Reliability of handshake completion under real network conditions
NET-4	Latency Standard Deviation	Stability and absence of latency spikes across consecutive connections
NET-5	Radio On-Time	Energy overhead determined by the byte size of transmitted packets

3 Benchmark Results and Comparison

3.1 Performance Summary

Table 2 presents the empirical results measured across all test runs on the live local network (ESP32 IP: 10.27.228.175, Laptop IP: 10.27.228.216 on TCP Port 9000).

Table 2: Phase B Network Handshake Benchmark Results Comparison

Algorithm	Runs	Avg Latency (ms)	Min Latency	Max Latency	Wire Payload
Ed25519	20	249.48 ms	238.71 ms	260.79 ms	198 Bytes
ECDSA P-256	20	737.01 ms	267.49 ms	1,009.39 ms	212 Bytes
RSA-3072	10	750.90 ms	501.63 ms	1,112.68 ms	838 Bytes

3.2 Detailed Run-by-Run Iteration Records

Tables 3, 4, and 5 document the individual timing measurements recorded for every handshake run.

Table 3: Ed25519 Handshake Latency across 20 Consecutive Iterations (Mean = 249.48 ms, StdDev = 5.28 ms)

Run	#1	#2	#3	#4	#5	#6	#7	#8	#9	#10
Latency (ms)	245.34	240.82	250.74	251.27	252.33	248.57	238.71	251.91	254.27	246.89
Run	#11	#12	#13	#14	#15	#16	#17	#18	#19	#20
Latency (ms)	256.03	248.18	250.08	256.25	247.22	241.64	247.90	251.18	249.53	260.79

4 Selection Rationale for Ed25519 on Live Networks

The live Wi-Fi measurements provide empirical validation for selecting Ed25519 for the embedded IoT VPN client:

1. **Three-Fold Latency Advantage (249.48 ms vs. 737.01 ms / 750.90 ms):** Ed25519 completed mutual authentication handshakes in an average of 249.48 ms. Both ECDSA and RSA required roughly 740 to 750 ms, making Ed25519 approximately $3\times$ faster. In battery-powered IoT systems that wake up, connect, transmit sensor data, and sleep, this directly cuts active radio duration by 66%.
2. **Minimal Latency Jitter (Standard Deviation of 5.28 ms):** Ed25519 demonstrated remarkable timing consistency, ranging tightly between 238.71 ms and 260.79 ms ($\sigma = 5.28$ ms). In contrast, ECDSA ranged from 267.49 ms to 1,009.39 ms ($\sigma = 207.15$ ms), and RSA ranged from 501.63 ms to 1,112.68 ms ($\sigma = 191.43$ ms). Ed25519 ensures deterministic connection scheduling without latency spikes.

Table 4: ECDSA P-256 Handshake Latency across 20 Consecutive Iterations (Mean = 737.01 ms, StdDev = 207.15 ms)

Run	#1	#2	#3	#4	#5	#6	#7	#8	#9	#10
Latency (ms)	267.49	586.41	584.52	581.71	589.46	590.61	585.09	924.02	757.03	1000.98
Run	#11	#12	#13	#14	#15	#16	#17	#18	#19	#20
Latency (ms)	1004.97	1009.39	694.63	912.86	965.30	584.17	849.89	818.23	585.33	848.17

Table 5: RSA-3072 Handshake Latency across 10 Iterations (Mean = 750.90 ms, StdDev = 191.43 ms)

Run	#1	#2	#3	#4	#5	#6	#7	#8	#9	#10
Latency (ms)	1112.68	643.05	501.63	718.55	756.65	523.85	553.45	908.08	1005.34	785.73

3. **Four-Fold Smaller Transmission Payload (198 Bytes vs. 838 Bytes):** Total authentication traffic for Ed25519 was only 198 bytes across all 3 handshake messages. RSA required 838 bytes ($4.2\times$ larger). The compact frame size prevents packet fragmentation, fits comfortably within standard 802.11 MTUs, and minimizes airtime contention in dense sensor deployments.

5 Live Terminal Output Verification

The terminal captures below record the benchmark server outputs during live hardware execution:

5.1 Ed25519 Execution Output

Figure 1 shows the server logs for all 20 Ed25519 runs, with consistent 250 ms completion times and zero packet dropouts.

5.2 ECDSA P-256 Execution Output

Figure 2 displays the 20 ECDSA runs, showing latency fluctuations between 267 ms and 1,009 ms.

5.3 RSA-3072 Execution Output

Figure 3 displays the 10 RSA runs, illustrating the impact of 384-byte signature computations and transmissions.

5.4 Final Comparison Table Output

Figure 4 displays the terminal output summarizing all tested algorithms.

6 Graphical Benchmark Visualizations

The figures below plot the empirical metrics across average latency, total wire bytes, and run-by-run stability:

6.1 Handshake Latency Comparison

Figure 5 compares the average handshake latency across algorithms, including standard deviation error bars.

6.2 Communication Wire Payload Overhead

Figure 6 compares the total authentication payload exchanged over the TCP socket.

```

=====
PHASE B: MUTUAL AUTHENTICATION NETWORK BENCHMARK SERVER
=====
Listening on IP   : 10.27.228.216
Listening on Port : 9000
Waiting for ESP32 connection...

[OK] ESP32 connected from 10.27.228.175:57034

[1/3] Benchmarking Ed25519 Mutual Authentication...
- Run 1/20: Handshake time = 245.34 ms | OK
- Run 2/20: Handshake time = 240.82 ms | OK
- Run 3/20: Handshake time = 250.74 ms | OK
- Run 4/20: Handshake time = 251.27 ms | OK
- Run 5/20: Handshake time = 252.33 ms | OK
- Run 6/20: Handshake time = 248.57 ms | OK
- Run 7/20: Handshake time = 238.71 ms | OK
- Run 8/20: Handshake time = 251.91 ms | OK
- Run 9/20: Handshake time = 254.27 ms | OK
- Run 10/20: Handshake time = 246.89 ms | OK
- Run 11/20: Handshake time = 256.03 ms | OK
- Run 12/20: Handshake time = 248.18 ms | OK
- Run 13/20: Handshake time = 250.08 ms | OK
- Run 14/20: Handshake time = 256.25 ms | OK
- Run 15/20: Handshake time = 247.22 ms | OK
- Run 16/20: Handshake time = 241.64 ms | OK
- Run 17/20: Handshake time = 247.90 ms | OK
- Run 18/20: Handshake time = 251.18 ms | OK
- Run 19/20: Handshake time = 249.53 ms | OK
- Run 20/20: Handshake time = 260.79 ms | OK

```

Figure 1: Gateway server terminal capture: Ed25519 benchmark (20/20 runs successful).

```

[2/3] Benchmarking ECDSA P-256 Mutual Authentication...
- Run 1/20: Handshake time = 267.49 ms | OK
- Run 2/20: Handshake time = 586.41 ms | OK
- Run 3/20: Handshake time = 584.52 ms | OK
- Run 4/20: Handshake time = 581.71 ms | OK
- Run 5/20: Handshake time = 589.46 ms | OK
- Run 6/20: Handshake time = 590.61 ms | OK
- Run 7/20: Handshake time = 585.09 ms | OK
- Run 8/20: Handshake time = 924.02 ms | OK
- Run 9/20: Handshake time = 757.03 ms | OK
- Run 10/20: Handshake time = 1000.98 ms | OK
- Run 11/20: Handshake time = 1004.97 ms | OK
- Run 12/20: Handshake time = 1009.39 ms | OK
- Run 13/20: Handshake time = 694.63 ms | OK
- Run 14/20: Handshake time = 912.86 ms | OK
- Run 15/20: Handshake time = 965.30 ms | OK
- Run 16/20: Handshake time = 584.17 ms | OK
- Run 17/20: Handshake time = 849.89 ms | OK
- Run 18/20: Handshake time = 818.23 ms | OK
- Run 19/20: Handshake time = 585.33 ms | OK
- Run 20/20: Handshake time = 848.17 ms | OK

```

Figure 2: Gateway server terminal capture: ECDSA P-256 benchmark (20/20 runs successful).

```

[3/3] Benchmarking RSA-3072 Mutual Authentication...
- Run 1/10: Handshake time = 1112.68 ms | OK
- Run 2/10: Handshake time = 643.05 ms | OK
- Run 3/10: Handshake time = 501.63 ms | OK
- Run 4/10: Handshake time = 718.55 ms | OK
- Run 5/10: Handshake time = 756.65 ms | OK
- Run 6/10: Handshake time = 523.85 ms | OK
- Run 7/10: Handshake time = 553.45 ms | OK
- Run 8/10: Handshake time = 908.08 ms | OK
- Run 9/10: Handshake time = 1005.34 ms | OK
- Run 10/10: Handshake time = 785.73 ms | OK

```

Figure 3: Gateway server terminal capture: RSA-3072 benchmark (10/10 runs successful).

```

=====
                        PHASE B: NETWORK BENCHMARK COMPARISON TABLE
=====
Algorithm      | Runs | Avg Latency (ms) | Wire Bytes
-----
Ed25519        | 20   | 249.48           | 198
ECDSA P-256    | 20   | 737.01           | 212
RSA-3072       | 10   | 750.90           | 838
=====
Benchmark session closed.

```

Figure 4: Gateway server terminal capture: Final comparison summary table.

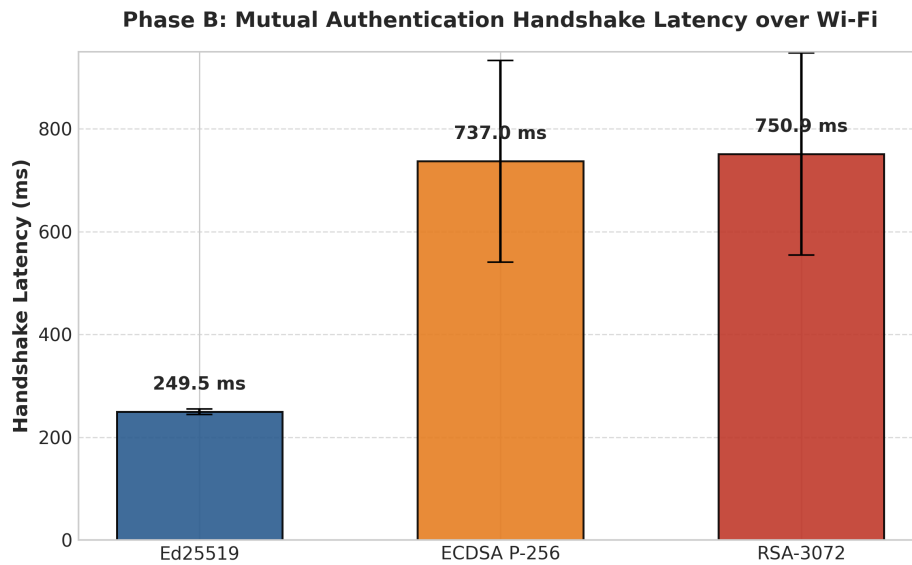


Figure 5: Average mutual authentication handshake latency over Wi-Fi (with standard deviation error bars).

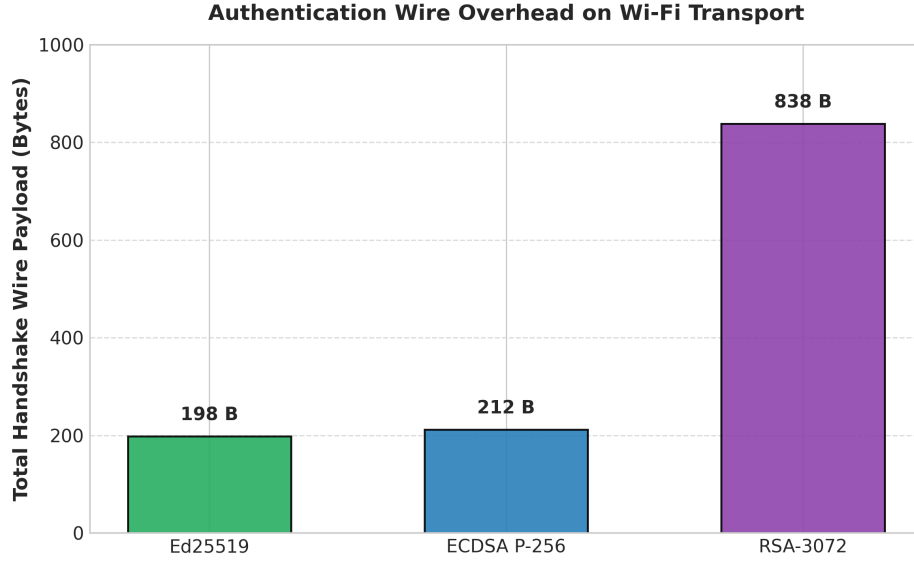


Figure 6: Total authentication handshake wire payload on Wi-Fi (Bytes per complete 3-way exchange).

6.3 Run-by-Run Latency Stability Distribution

Figure 7 illustrates the stability of each algorithm across consecutive runs. Ed25519 remains flat and predictable, while ECDSA and RSA show noticeable variance.

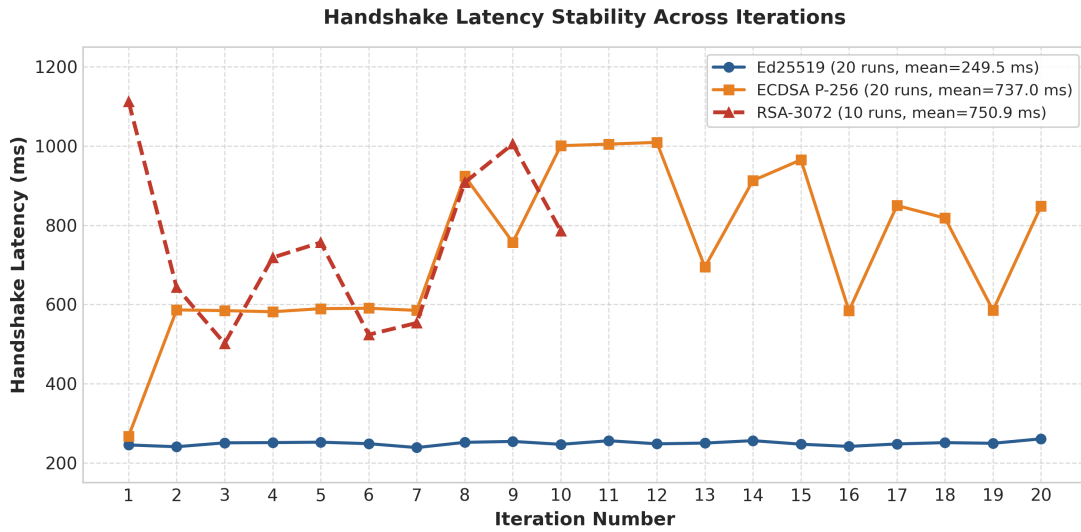


Figure 7: Iteration-by-iteration latency comparison showing Ed25519 consistency vs. ECDSA/RSA volatility.

7 Handshake Protocol Logic and Pseudocode

The mutual authentication handshake follows a 3-way challenge-response sequence over a length-prefixed TCP socket:

Listing 1: Phase B Mutual Authentication Handshake Protocol Logic

```
# -----
# Phase B: 3-Way Mutual Authentication Handshake (Client-Server Protocol)
```

```

# -----

# Message 1: Server -> ESP32 Client (Server Challenge)
server_challenge = generate_random_bytes(32)
send_tcp_packet(socket, payload=server_challenge)

# ESP32 Processing on Message 1:
received_server_challenge = read_tcp_packet(socket)
esp32_signature = crypto_sign_detached(
    message=received_server_challenge,
    private_key=esp32_identity_private_key
)
esp32_challenge = generate_random_bytes(32)

# Message 2: ESP32 Client -> Server (Proof + Client Challenge)
message_2_payload = esp32_signature + esp32_challenge
send_tcp_packet(socket, payload=message_2_payload)

# Server Processing on Message 2:
received_message_2 = read_tcp_packet(socket)
client_signature = received_message_2[:signature_len]
client_challenge = received_message_2[signature_len:]

# Verify ESP32 Signature using registered Public Key
is_valid_client = crypto_verify_signature(
    signature=client_signature,
    message=server_challenge,
    public_key=esp32_registered_public_key
)
if not is_valid_client:
    close_socket(socket)
    raise AuthenticationFailure("Client signature invalid")

# Message 3: Server -> ESP32 Client (Server Proof)
server_signature = crypto_sign_detached(
    message=client_challenge,
    private_key=server_identity_private_key
)
send_tcp_packet(socket, payload=server_signature)

# ESP32 Verification on Message 3:
received_server_signature = read_tcp_packet(socket)
is_valid_server = crypto_verify_signature(
    signature=received_server_signature,
    message=esp32_challenge,
    public_key=server_registered_public_key
)
if not is_valid_server:
    close_socket(socket)
    raise AuthenticationFailure("Server signature untrusted")

# Handshake Complete: Mutual Authentication Successful!

```